EXO-INSIGHT

AI Assistance Log and Conversation Summary
Coding for AI & Data Science — PGE3

Purpose. This document summarizes AI assistance used during the EXO-INSIGHT project, based on project context and conversation details available in this session.

Important: This is an AI-prepared summary, not a verbatim export of complete ChatGPT conversations. Attach original ChatGPT conversation PDFs separately if the assignment requires conversation exports.

1. Project context and AI role

EXO-INSIGHT is an exoplanet screening and machine-learning project using the NASA Exoplanet Archive PSCompPars dataset. Work covered data profiling, preprocessing, scientific screening, candidate analysis, decision modelling, local language-model explanations, benchmarking and uncertainty analysis.

AI assistance focused on Python and PowerShell debugging, reviewing implementation outputs, explaining evaluation results, preparing tests and documentation, and identifying scientific wording risks. The student ran commands locally, reviewed outputs and decided whether to apply proposed changes.

2. M6 — Local model integration and explanation

- Reviewed the M6 implementation and local Qwen integration through Ollama.
- Diagnosed truncated explanations and suggested increasing num_predict from 220 to 400 tokens.
- Checked that the numerical score was called a label-match score, not model confidence.
- Identified generated wording that incorrectly described planetary mass as host-star mass and misdescribed the mass-provenance field.
- Suggested explicit feature-unit metadata and prompt safeguards for planetary mass, radius and mass provenance.

Student execution and observed result: The student applied the changes and ran the end-to-end command for 11 Com b. The displayed result included prediction label 1, label-match score 0.915869, threshold 0.50, planetary radius 12.2 Earth radii, planetary mass 4,914.89848585 Earth masses, discovery year 2007, mass-provenance value 'Msini', and discovery method Radial Velocity. The final Qwen explanation correctly described planetary mass and radius, preserved 'Msini' without expansion, and stated that the output does not establish habitability or life.

Corrections and rationale: The output-token limit was increased because the initial explanation was truncated. Explicit units were added because the explanation initially called the mass unit unspecified. Prompt safeguards were added after Qwen incorrectly called the planet's mass the host star's mass and misdescribed the mass-provenance field. These changes did not alter the trained classifier, its five model features, prediction or score.

3. M7 — Model benchmarking

AI assistance helped implement and interpret a benchmark comparing the EXO-INSIGHT classifier with Qwen used alone on ten held-out labelled cases, with three runs per case (30 Qwen calls). All 30 Qwen responses were valid, but Qwen predicted class 1 for every case. The classifier achieved 0.90 accuracy in each run; Qwen-alone achieved 0.50. Both standard deviations were zero under the deterministic configuration (temperature 0).

A trace showed one Qwen error: HD 83443 c had benchmark label 0, while Qwen returned 1. The trace supports reporting the incorrect prediction and its input features, but not a causal explanation. The benchmark is exploratory because it uses only ten cases. Artifacts: artifacts/m7_benchmark_summary.json and artifacts/m7_benchmark_traces.csv.

4. M8 — Uncertainty analysis

AI assistance helped implement and interpret bootstrap analysis on the held-out test set using 2,000 resamples and random seed 42. The test set contained 237 rows: 18 positive and 219 negative.

Metric	Observed	95% bootstrap interval
Accuracy	0.7004	[0.6414, 0.7595]
Balanced accuracy	0.8124	[0.7422, 0.8641]

The intervals quantify sampling uncertainty against M4 protocol-derived pseudo-labels, not uncertainty about actual planetary habitability. The small positive class (18 cases) limits the strength of conclusions. Artifact: artifacts/m8_uncertainty_results.json.

5. Testing, reproducibility and version control

AI assistance helped configure the test import path, expand M6 tests and interpret failures. The student ran the final test suite locally: 11 tests passed in 5.84 seconds using Python 3.10.11 and pytest 9.2.0.

README.md, EVIDENCE.md, requirements.txt and AI LOG/SESSION_NOTES.md were created or reviewed during the work. Git was initialized and a local commit was created: 35a273a — 'Complete EXO-INSIGHT final project implementation'. The reported working tree was clean immediately after the commit.

6. Student review and academic integrity

The student must review this summary before submission, retain only statements that accurately describe their own actions and understanding, add any missing AI-use sessions, and attach original conversation exports where required. The student should be prepared to explain the code, scientific assumptions, model evaluation, benchmark design, bootstrap intervals and limitations independently.

Suggested original conversation PDFs to accompany this log: M6 implementation, M7 benchmarking, M8 uncertainty analysis, and earlier sessions materially used for data profiling, preprocessing, scientific validation or M5 modelling.

Prepared from available project conversation context on 3 November 2026. This summary is not a substitute for original ChatGPT conversation exports or the student's own reflective notes.